

Topic	Construction of robust predictive models to ensure reliable generalisation to new and unseen observations.				
Title	Link	Authors	Citations	Date of Publish	Key Paper
A guide to machine learning for biologists	https://www.nature.com/articles/s41580-021-00407-0	Greener J.G., Kandathil S.M., Moffat L., Jones D.T.	1900	2021	No
Evaluating generalizability of artificial intelligence models for molecular datasets	https://www.nature.com/articles/s42256-024-00931-6	Ektefaie Y., Shen A., Bykova D., Marin M.G., Zitnik M., Farhat M.	27	2024	No
Machine learning models and over-fitting considerations	https://pmc.ncbi.nlm.nih.gov/articles/PMC8905023/	Charilaou P., Battat R.	169	2022	No
Optimizing for Generalization in Machine Learning with Cross-Validation Gradients	https://pmc.ncbi.nlm.nih.gov/articles/PMC8905023/	-	-	-	No
Pitfalls and Remedies for Cross Validation with Multi-trait Genomic Prediction Methods	https://pmc.ncbi.nlm.nih.gov/articles/PMC6829121/	Runcie D., Cheng H.	-	2019	No
Regularized Machine Learning in the Genetic Prediction of Complex Traits	https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1004754	Okser S., Pahikkala T., Airola A., Salakoski T., Ripatti S., Aittokallio T.	158	2014	No
Unsupervised domain adaptation methods for cross-species transfer of regulatory code signals	https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2023.1140663/full	Latyshev P., Pavlov F., Herbert A., Poptsova M.	-	2023	No
Exploration, Inference, and Prediction in Neuroscience and Biomedicine	https://www.cell.com/trends/neurosciences/fulltext/S0166-2236(19)30007-4	Bzdok D., Meyer-Lindenberg A.	180	2019	No
Enhancing scientific discoveries in molecular biology with deep generative models	https://link.springer.com/article/10.15252/msb.20199198	Lopez R., Gayoso A., Yosef N.	85	2020	No
Illusory generalizability of clinical prediction models	https://www.science.org/doi/10.1126/science.adg8538	Chekroud A.M. et al.	251	2024	Yes
Cross-trial prediction of treatment outcome in depression: a machine learning approach	https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366(15)00471-X/fulltext	Chekroud A.M. et al.	571	2016	No
Predictive biology: modelling, understanding and harnessing microbial complexity	https://www.nature.com/articles/s41579-020-0372-5	Lopatkin A.J., Collins J.J.	164	2020	No
Predictive models avoid excessive reductionism in cognitive neuroimaging	https://www.sciencedirect.com/science/article/pii/S0959438818301089	Varoquaux G., Poldrack R.A.	48	2019	No
Current challenges for unseen-epitope TCR interaction prediction	https://academic.oup.com/bib/article/22/4/bbaa318/6042663	Moris P. et al.	204	2020	Yes
How evolution learns to generalise	https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005358	Kouvaris K. et al.	96	2017	No
Improving generalization of machine learning-identified biomarkers using causal modelling	https://www.nature.com/articles/s42256-023-00781-8	Pavlović M. et al.	17	2024	Yes
Beyond generalization: a theory of robustness in machine learning	https://link.springer.com/article/10.1007/s11229-023-04334-9	Freiesleben T., Grote T.	97	2023	No
Reliable prediction of amino acid interactions in enzyme evolution	https://www.nature.com/articles/s41598-018-35033-y	Cadet F. et al.	135	2018	No
Making predictive modelling ART: accurate, reliable, and transparent	https://esajournals.onlinelibrary.wiley.com/doi/epdf/10.1002/ecs2.3160	Bodner K.O., Fortin M.J., Molnár P.K.	31	2020	No
Comparison of statistical methods commonly used in predictive modelling	https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1654-1103.2004.tb02263.x	Muñoz J., Felicísimo Á.M.	99	2004	No

Transfer learning enhanced GANs for MRI reconstruction	https://www.sciencedirect.com/science/article/pii/S0010482521002985	Lv J. et al.	68	2021	No
Preventing dataset shift from breaking machine-learning biomarkers	https://academic.oup.com/gigascience/article/10/9/giab055/6377530	Dockès J., Varoquaux G., Poline J.B.	104	2021	Yes
Invariant Risk Minimization	https://arxiv.org/pdf/1907.02893	Arjovsky M., Bottou L., Gulrajani I., Lopez-Paz D.	3359	2020	No
How to avoid machine learning pitfalls: a guide for academic researchers	https://arxiv.org/html/2108.02497v4	Lones M.A.	161	2024	No
Navigating the pitfalls of applying machine learning in genomics	https://www.nature.com/articles/s41576-021-00434-9	Whalen S., Schreiber J., Noble W.S., Pollard K.S.	290	2021	Yes
TRIPOD+AI statement: reporting clinical prediction models	https://www.bmj.com/content/385/bmj-2023-078378	Collins G.S. et al.	2189	2024	Yes